

Adaptive Segmentation Enhanced Asynchronous Federated Learning for Sustainable Intelligent Transportation Systems

Xiaokang Zhou¹, *Member, IEEE*, Wei Liang², *Member, IEEE*, Akira Kawai, *Member, IEEE*,
Kaoru Fueda, Jinhua She³, *Fellow, IEEE*, and Kevin I-Kai Wang⁴, *Member, IEEE*

Abstract—The proliferation of advanced embedded and communication technologies has facilitated the possibility of modern Intelligent Transportation System (ITS). The hierarchical nature of such large-scale and distributed systems brings obvious challenges in creating a scalable and sustainable computing environment, and hence the development and application of edge intelligence become critical. Federated learning (FL), as an emerging distributed machine learning paradigm, aims to offer secure knowledge sharing and effective learning across multiple devices. However, conventional FL may fall into trouble when facing large-scale and network-agnostic systems with fast moving devices and changing network attributes. In this study, we propose an Adaptive Segmentation enhanced Asynchronous Federated Learning (AS-AFL) model, aiming to improve the learning efficiency and reliability in sustainable ITS via a decentralized fashion. Specifically, a meta-learning based adaptive segmentation scheme is designed to automatically separate the client nodes (e.g., vehicles) into multiple edge groups according to their homogeneous attributes. An integrated aggregation mechanism is then developed to realize the horizontal FL among a group of similar client nodes via the so-called intra-group synchronous aggregation, while allowing the vertical FL across different groups via the so-called inter-group asynchronous aggregation. Experiment and evaluation results based on an open-source dataset demonstrate the outstanding learning and communication performance of our proposed model, compared with several conventional FL schemes in a distributed ITS application scenario.

Index Terms—Decentralized federated learning, adaptive segmentation, sustainable computing, intelligent transportation systems.

I. INTRODUCTION

THE wide penetration and deployment of communication infrastructures and embedded devices are moving the future Internet of Things (IoT) technologies beyond personal applications. Geographically distributed and hierarchical IoT systems for applications, such as smart cities/nations, pervasive healthcare, and Intelligent Transportation Systems (ITS), are soon becoming the reality [1], [2], [3], [4]. Large amounts of data are expected to be generated by billions of interconnected devices that are communicating via different networks and different protocols [5]. The hierarchical nature of such large-scale systems brings obvious challenges in managing scalability and sustainability. Edge intelligence is becoming a critical aspect to ensure scalable and reliable distributed systems in future sustainable ITS, which calls for new distributed machine learning techniques to effectively manage and leverage the large amounts of data collected over heterogeneous devices and networks across hierarchical topologies [6].

It is noticed that traditional centralized machine learning which requires data to be transmitted to a central location is no longer suitable for such hierarchical and distributed networks and applications. To ensure effective use of the distributed data and resources within particular edge groups, distributed machine learning schemes are developed for the purpose of offering distributed edge intelligent services. In particular, Federated Learning (FL) is an emerging distributed machine learning paradigm that aims to achieve a more efficient and accurate distributed learning through effective and secure knowledge sharing [7], [8].

Despite the tremendous research effort, the heterogeneity and scalability issues in hierarchical ITS are still posing significant challenges to existing distributed machine learning techniques. Taking a typical scenario as an example, different vehicular networks may exist in different parts of a transportation system (e.g., V2V, V2I, or V2X), while existing centralized FL frameworks require a central server to collect model parameters from those individual vehicles and perform model aggregation. This centralized server brings scalability

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Xiaokang Zhou is with the Faculty of Data Science, Shiga University, Hikone 522-8522, Japan, and also with the RIKEN Center for Advanced Intelligence Project, RIKEN, Tokyo 103-0027, Japan (e-mail: zhou@biwako.shiga-u.ac.jp).

Wei Liang is with the School of Advanced Interdisciplinary Studies and School of Management Science and Engineering, Hunan University of Technology and Business, Changsha 410205, China, and also the Xiangjiang Laboratory, Changsha 410205, China (e-mail: weilang@csu.edu.cn).

Akira Kawai and Kaoru Fueda are with the Faculty of Data Science, Shiga University, Hikone 522-8522, Japan (e-mail: akira-kawai@biwako.shiga-u.ac.jp; kfueda@biwako.shiga-u.ac.jp).

Jinhua She is with the School of Engineering, Tokyo University of Technology, Tokyo 192-0982, Japan (e-mail: she@stf.teu.ac.jp).

Kevin I-Kai Wang is with the Department of Electrical, Computer, and Software Engineering, University of Auckland, Auckland 1010, New Zealand (e-mail: kevin.wang@auckland.ac.nz).

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and sustainability issue for the targeted distributed and network-agnostic transportation system since existing practical vehicular networks do not have guaranteed stable connection or communication latency. It becomes a challenging issue to achieve the synchronized parameter upload from multiple local clients for a centralized model aggregation. The fact that all the parameters are collected by a central server during the model aggregation also brings the risk of a single point of failure, which is highly vulnerable to malicious attacks, especially in sustainable computing systems. Therefore, both the nature of heterogeneous network topology and the formation of multiple edge computing groups need to be considered, when addressing issues of network-agnostic scalability and sustainability in smart application development.

In this study, to cope with the problems discussed above, an Adaptive Segmentation enhanced Asynchronous Federated Learning (AS-AFL) model is proposed and designed. Different to the traditional centralized FL, the proposed framework implements FL in a decentralized fashion to achieve edge intelligence in sustainable computing systems, where the model aggregation is performed in a peer-to-peer manner, without having a central server. A meta-learning scheme is developed and utilized to separate the whole local clients into multiple edge groups automatically, according to their various network attributes. Given an ITS application scenario for example, individual vehicles within one edge group can perform parameter update and aggregation synchronously in a peer-to-peer manner, while the individual group may receive the updated model parameters from its neighboring groups to realize the model aggregation asynchronously across multiple groups, accommodating different network performance and heterogeneity. Main contributions can be summarized as follows.

- i) A hybrid FL framework is constructed to enhance the efficiency and reliability in sustainable ITS, which includes a decentralized horizontal FL scheme among multiple vehicles via a peer-to-peer manner, and a vertical FL scheme that ensures secure knowledge/model sharing across different groups.
- ii) A meta-learning based adaptive segmentation scheme is designed to enable the automatic edge group formation considering the multiple factor such as client attributes, metadata, network attributes, and types of training models, which can significantly improve the FL performance especially in large-scale and network-agnostic ITS application scenarios.
- iii) An integrated aggregation mechanism is developed, which allows the intra-group synchronous aggregation within a group of similar client nodes to ensure the integrity of training even under typical vehicular networks with potentially high latency, while enables the inter-group asynchronous aggregation from a series of groups to effectively alleviate the delay especially in complex and resource-limited communication environment.

The rest of this article is organized as follows. Section II provides an overview of related works on applications of FL targeting distributed system with edge intelligence. The

problem definition and basic framework are presented in Section III, followed by the detailed mechanism and algorithm implemented in the proposed model in Section IV. Section V discusses the experiment design and evaluation results in comparison with conventional FL approaches. We conclude this study and give our promising perspectives regarding future research in Section VI.

II. RELATED WORK

In this section, current studies related to FL techniques and edge intelligence applications in distributed systems, are summarized and discussed.

A. FL for Distributed Intelligence Development

Currently, FL has been viewed as a promising privacy-preserving learning technique to deploy distributed intelligence with various IoT devices in end-edge-cloud based sustainable computing systems. Duan et al. [9] introduced a so-called self-balancing FL framework, to deal with the imbalanced and distributed data training in mobile or IoT systems. They re-scheduled the CNN-based local training using Kullback–Leibler divergence, and utilized a Z-score-based data augmentation method to alleviate the global imbalance. Aiming at providing diverse quality-of-services in future IoT networks, Messaoud et al. [10] proposed a federated reinforcement learning framework, in which a multi-agent federated Q-learning scheme was utilized to build the global model for dynamic network management and resource allocation. Wang et al. [11] gave a theoretical analysis on gradient-descent based FL model in resource constrained edge computing systems. They developed a control algorithm to learn the data distribution from multiple edge nodes, which could minimize the loss function and optimize the frequency of global aggregation considering the system dynamics. Chen et al. [12] presented the FL in an asynchronous strategy, which enabled parameters in shallow and deep layers to be aggregated and updated differently in the local model. They then employed a temporally weighted aggregation scheme in the server to enhance the convergence of the global model. Hao et al. [13] designed a non-interactive FL scheme for industrial IoT development. They embedded a homomorphic ciphertext based on private gradients in the augmented learning, and employed a distributed Gaussian mechanism to enhance privacy protection during the local training process. Sattler et al. [14] extended a compression method to enhance the communication efficiency in FL. Their proposed communication-efficient FL scheme could compress both upstream and downstream communications, and optimize the encoding of weight update. Mowla et al. [15] proposed an adaptive FL architecture with reinforcement learning for on-vehicle jamming attack detection, in which they used input from the FL model to update the Q-table, and employed the adaptive epsilon greedy policy in Q-learning to optimize the defense path. Qu et al. [16] focused on a decentralized paradigm for big data driven cognitive computing in Industry 4.0 networks based on the integration of FL and blockchain techniques. They further developed

a Markov based optimization mechanism to improve the model performance against poisoning attacks. Considering the device-level and data-level heterogeneity in federated mobile devices, Wang et al. [17] exploited two near-optimal algorithms for different workload schedules when dealing with identically or non-identically distributed data.

B. Edge Intelligence in Distributed Applications

With the increasing popularity of IoT, more and more distributed data is generated by widespread mobile and IoT devices, which calls for edge intelligence with deep learning empowered analytics in AI-based IoT applications. Deng et al. [18] discussed the integration of edge computing and AI techniques, i.e., edge intelligence in distributed systems, and introduced several examples on how to strengthen edge with intelligence in terms of framework design, processor acceleration, and model adaptation. Wang et al. [19] investigated deep learning techniques for edge intelligence based on a series of application scenarios. They further discussed several fundamental issues, including communication, computing power, energy consumption, in realizing the deployment of practical deep learning services in edge computing frameworks. Zhou et al. [20] applied the FL framework in distributed recommender system, in order to provide personalized and context-aware recommendations in a big data scenario. Their theoretical analysis indicated that the integration of FL could provide a privacy-preserving model training with lower latency and less power consumption. Zhang et al. [21] presented a so-called multiple algorithm service model, which could be deployed with heterogeneous algorithms for different data sizes with different computation complexities. They also considered an energy-delay optimization model with a tide ebb algorithm for the energy consumption issue in edge AI applications. Chiu et al. [22] designed a FL-based edge learning system, in which a semi-supervised distributed learning scheme was developed to train local models in different edge devices, and a so-called federated swapping operation was introduced to deal with weight divergence among the distributed data. Utkarsh et al. [23] built a consensus-based dimension-distributed computational intelligence framework, which could minimize the communication overhead for real-time optimization control in smart distribution grids. Dong et al. [24] focused on distributed intelligence in Internet of Vehicles, and addressed a distributed machine learning enhanced intelligent driving decision model. They modeled the worker selection problem in a deep neural network, and used a reinforcement learning method to optimize the policy network in a Markov decision process. Ioannou et al. [25] aimed to utilize the distributed intelligent technique to control the device-to-device network generation. To this purpose, they developed a proof-of-concept algorithm to better use the intelligent agents when selecting transmission modes with lower computational load in 5G networks. Shi et al. [26] constructed a distributed fuzzy neural network, to deal with the uncertainty and distributed challenges in big data environments. They further proposed a consensus learning algorithm, which could realize the structure learning and

parameter learning based on alternating direction method of multipliers.

III. AS-AFL FRAMEWORK

In this section, we first briefly introduce the framework of our proposed model, then give a series of formal definitions in the ITS application scenario.

A. Application Scenario and Basic Framework

Considering a network-agnostic ITS environment with many different vehicles, each vehicle may have different attributes and computing capabilities, and generate different raw data with models for different kinds of tasks from the heterogeneous sub-networks. In such situation, the transferred parameters may not be of the same type, thus we are not able to share data directly, and build a large number of models to integrate the parameters, due to the huge consumption of computing and communication resources. In addition, the low probability of abnormal vehicle activities may lead to a lack of well-labeled data for model training, while conventional supervised machine learning or deep learning techniques require a large amount of labeled data, which makes the real-time monitoring or anomaly detection tasks more complex and challenging in modern ITS environments.

Therefore, an AS-AFL framework is designed to cope with the above-mentioned scalability and sustainability issues in a distributed ITS environment. Different from conventional horizontal or vertical FL models, the proposed AS-AFL conducts parameter aggregation via a peer-to-peer manner instead of on a central server. The detailed architecture is illustrated in Fig. 1.

As shown in Fig. 1, the proposed framework is basically composed of an adaptive segmentation module and an asynchronous FL module. First, a meta-learning based adaptive segmentation scheme is designed to separate all the client nodes (i.e., vehicles) into several groups according to their attributes, which can enhance the efficiency and reliability during the learning process. After training on different tasks, the meta-learning model may adapt and be generalized well to a new task, and learn to “learning to learn”. The model or learner is trained on a series of tasks with heterogeneous network conditions and training models during this phase, which enables the trained model to quickly adapt to the complex transportation environment, and separate the homogeneous vehicles into the same group for different tasks. A hybrid FL model associated with the synchronous and asynchronous aggregations is then applied across the distributed nodes. Differing to conventional centralized vertical or horizontal FL scheme, parameters learned in each client will be aggregated locally by a peer-to-peer manner within a group. The end devices (i.e., the vehicles) periodically query the neighboring devices and exchange the learning parameters directly through the Gossip protocol in a distributed aggregation strategy. The so-called intra-group synchronous aggregation may enable the clients with identical group label to be treated as homogeneous nodes, so as to achieve an efficient synchronous aggregation within the same

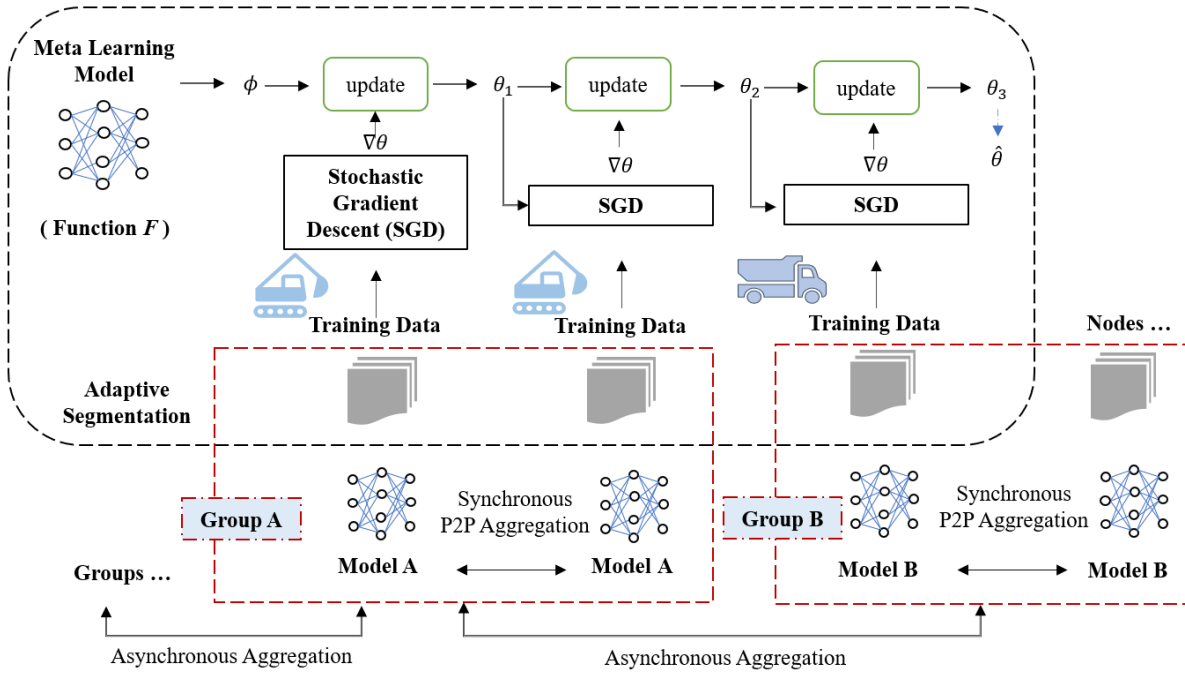


Fig. 1. AS-AFL framework architecture.

group among similar devices following a meta-learning based segmentation scheme. On the other hand, the so-called inter-group asynchronous aggregation is more secure and robust across different groups because the framework has no single point of failure, which can effectively mitigate the network requirements as much as possible in the complex and resource-limited communication environment, especially in case of the connection or central node failure.

B. Model Definitions

Given a client node set $D = \{d_1, d_2, \dots, d_N\}$ representing N nodes (i.e., vehicles) in a complex distributed ITS. The data generated by D is defined as $X = \{x_1, x_2, \dots, x_N\}$, and $Y = \{y_1, y_2, \dots, y_N\}$ denotes its corresponding label set. Client nodes distributed in various regions have different attributes, according to different conducted tasks. The adaptive meta-learning scheme may learn an intelligent strategy that can efficiently classify the clients to different tasks. Accordingly, each data type of the dataset may not be exactly the same, which can be divided into M groups according to the specific categories, and the labels of the groups are set to $g^{(1)}, g^{(2)}, \dots, g^{(M)}$. A segmentation task may contain one or more groups of labels, while each task has K data samples. For example, given the group label as $g^{(1)}$ after grouping based on the attributes, the data in the label can be described as $Attr = \{attr_1, attr_2, \dots, attr_K\}$. Based on the above definitions, a segmented LAN LAN_m of the investigated dynamic environment of m -th group can be defined as follows.

$$LAN_m = \{EN_m, Meta_m, trie_\Delta\} \quad (1)$$

where EN_m indicates all the online nodes within LAN_m , which will be dynamically organized by the adaptive segmentation and heart-beating signals. $Meta_m$ indicates the

hyperparameters, including the maximum count of LAN nodes, learning terminating threshold and timeout, and meta-learning hyperparameters. $trie_\Delta$ denotes the set of network status tree snapshots of LAN_m , and Δ denotes the updated gradient.

Assuming each task $T^{(m)} = \{LAN_m, Loss_m, x_m, y_m\}$ stands for a trained model on a specific network environment with the objective function $Loss_m$ and training sample x_m, y_m . A meta-learning mechanism $A(*)$ is designed to maintain the $EN_m, Meta_m, trie_\Delta$ for LAN_m , and can be described as $A(T^{(1)}, T^{(2)}, \dots, T^{(M)}) \rightarrow F_\theta$, θ indicates the adaptive segmentation parameter and can be viewed as the meta-knowledge obtained in the learning process. Specifically, the M tasks: $T^{(1)}, T^{(2)}, \dots, T^{(M)}$ are sent into $A(*)$ for training, to obtain an optimal θ for multiple tasks and an algorithm F_θ . Formally, the goal of this study is to design an improved meta-learning scheme $A(*)$, considering how the average test error on new data changes across the heterogeneous networks during the adaptive segmentation.

IV. ASYNCHRONOUS FL FOR SUSTAINABLE ITS

In this section, we introduce the adaptive segmentation method based on a meta-learning scheme, and then discuss the integrated aggregation mechanism for the sustainable ITS.

A. Meta-Learning Based Adaptive Segmentation

We classify those tasks according to the attributes of individual vehicles. All vehicles along with their dataset are separated into M groups of tasks, and each task within M different directions owns at least K samples, which is call M -ways K -shot tasks. Meta-learning training is performed to get the initialized training parameters.

The designed adaptive segmentation scheme aims to find the attributes that are sensitive to affect both the efficiency and test error for different training tasks. The meta-learning mechanism $A(T^{(1)}, T^{(2)}, \dots, T^{(M)}) \rightarrow F_\theta$ applied in the segmentation scheme tries to detect the changes in the parameters which can produce large improvements on the loss function of any task. The meta-learning scheme applied in this study adopts a double-layer loop design in the learning phase. From the perspective of inner and outer loops of segmentation training process, in the inner loop, θ is obtained from the initialized parameter θ'_i of the previous round, trained by the Support set, with index i indicating that it has been trained by the i -th set of tasks. In the outer loop, θ'_i of the inner loop is evaluated using the total loss obtained from the Query set, following which θ is updated.

The initialized algorithm is obtained by putting tasks given by the Support set from the training data into the meta-learning mechanism $A(*)$ for training. Then the task given by the Query set in training data is involved to evaluate the meta-knowledge ω , calculate its loss function and gradient, and sum up its loss to get the total loss. The formulation can be described as follows.

$$L(\phi) = \sum_{m=1}^M loss^m \quad (2)$$

The calculated total loss is then used to evaluate θ , and the parameters are updated iteratively. During the training process, the initial parameters ϕ_0 first performs parameter optimization using a Support set of K -samples from the training task to obtain an optimized set of parameters, and then the Query set of the same set of training tasks is used to calculate the loss of that task and the gradient of the parameters as well. The parameters are updated as $\theta_m = \theta_m - \alpha \nabla_{\theta_m} \ell_{T^{(m)}}$, and after repeating the M sets of tasks, the parameters are fine-tuned using the Support set of test tasks, and the effect of meta-learning is finally evaluated using the Query set of test tasks. The adaptive segmentation based on meta-learning scheme is described in Algorithm 1.

As shown in Algorithm 1, a pruning strategy for adaptive segmentation can be obtained. First, the obtained gradients Δ stored in $trie_\Delta$ will be weighted temporally. Second, each pair of $trie_\Delta(m)$ and $\min(trie_\Delta)$ will be compared efficiently to evaluate and generate the differentiate sub-tree for $trie_\Delta(m)$ and $\min(trie_\Delta)$. Thus, pruning will be conducted to mitigate the workload for the whole system and optimize the searching efficiency, which makes it possible to learn and adapt quickly from just a few examples and continue to adapt as more data becomes available to achieve the adaptive meta-learning based grouping.

When the concrete segmentation strategy is learned by the meta-learning training process, the nodes will be re-organized into sub-groups. To explain the adaptive segmentation more clearly, an example of dynamic auto-group formation is given by Fig. 2. Considering four types of actions, namely Ping, FindNode, Neighbor, and Pong in this process, the strategy can discover new nodes by searching multiple disjoint paths in parallel, which use implicit ciphers to limit the generation of free vehicle ID, and introduce the reliable K_bucket (a routing table used to store node's NodeId, distance between

Algorithm 1 Adaptive Segmentation Based on Meta-Learning

Input: A set of tasks associated with heterogeneous network conditions and models $T^{(m)} = \{LAN_m, Loss_m, x_m, y_m\}_{m=1}^M$
Output: Updated $\{\theta_m\}$ obtained in the learning process $A(T^{(1)}, T^{(2)}, \dots, T^{(M)}) \rightarrow F_\omega$

```

1: Initialize parameter  $\{\theta_m\}$  for training tasks  $T^{(m)}$ 
2: Initialize the pruning tree  $trie_\Delta$ 
3: while not convergent do
4:   for  $m = 1$  to  $M$  do
5:     compute  $\nabla_{\theta_m} \ell_{T^{(m)}}(\theta_m, x_m, y_m)$  with respect to
        $K$  samples in task  $T^{(m)}$ 
6:     store the  $\nabla_{\theta_m}$  in the pruning tree as  $trie_\Delta(m)$ 
7:     conduct pruning for each pair of  $trie_\Delta(m)$  and
        $\min(trie_\Delta)$ 
8:     compute and update adaptive parameter  $\theta_m = \theta_m -$ 
        $\alpha \nabla_{\theta_m} \ell_{T^{(m)}}$ 
9:   end for
10: end while
11: return the learned model  $\{\theta_m\}$ 

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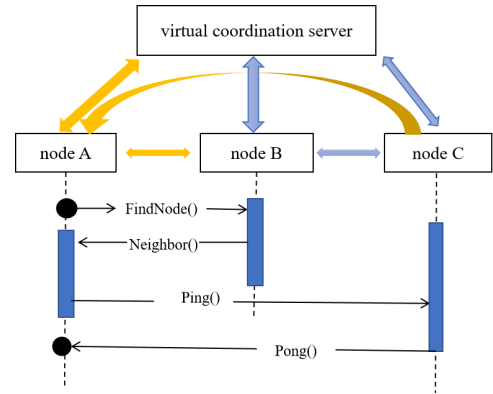


Fig. 2. Automatic discovery of new nodes during segmentation process.

two nodes, target nodes, etc.). As shown in Fig. 2, node A first sends a FindNode request to the nearby node B based on the K_bucket table, which indicates that it wants to query neighboring nodes of B. When the node B receives the FindNode message, it will send a Neighbor message to A in a timely manner, which contains the information about all neighboring nodes of B. Then, node A will add the information to its node list and send a Ping message to node C in the list. Node C will return a Pong message to A to prove that it is available. Finally, node A can connect to all nodes of the virtual coordination server, i.e., node A joins the group.

B. Asynchronous Aggregation in FL

Implementing adaptive meta-learning grouping can further improve FL and enable more efficient sharing and communication between vehicles. Vehicles with an identical group label are treated as homogeneous nodes, which will be trained under the intra-group synchronous aggregation. While the inter-group nodes will be aggregated via an asynchronous manner. In our problem definition, we consider a node set

$D = \{d_1, d_2, \dots, d_N\}$ with N samples defined as $\{x_n, y_n\}_{n=1}^N$ to indicate the data generated by D , in which x_n refers to the feature vector maintained by vehicle d_n , with a label y_n that is stored in the vehicle.

Based on the segmentation results discussed above, as the vehicle data $\{x_n, y_n\}_{n=1}^N$ is not shared with each other, to ensure the privacy of data, each vehicle is trained in a local model and the local parameter can be expressed as $\{\theta_n\}$ for $n = 1, 2, \dots, N$. An embedding h is introduced to map the original data x_n to the embedded information h_n , represent as $h_n := h(\theta_n, x_n)$.

Since there is no need for any centralized server to aggregate models in this framework, vehicles directly aggregate their local models within the shared information. Given a pair of vehicles $\{d_j, d_k\}$, $j, k = 1, 2, \dots, N$, when they are activated and have completed their local training, it can be described as:

$$F(\theta_j, \theta_k) := \frac{1}{2} \sum_{n=j,k} \ell(\theta_j, h_n; y_n) + r(\theta_k) \quad (3)$$

where $h_j := h(\theta_j, x_j)$ and $h_k := h(\theta_k, x_k)$. ℓ is the loss to capture the accuracy of the local parameter θ_j , and r is a regularization term.

A vertical FL scheme is considered in the asynchronous aggregation process since attributes of the local training models in the nodes are heterogeneous across different sub-networks. Given M vehicle groups that are activated and have completed their local training at one time, Algorithm 2 demonstrates the training process of vertical asynchronous FL across these groups.

Algorithm 2 Vertical Asynchronous FL Training Process

Input: A set of query samples $Q = \{x_m, y_m\}_{m=1}^M$

Output: Trained FL model with updated $\{\theta_m\}$

```

1: Initialize parameter  $\{\theta_m\}$ , for  $M$  vehicle groups
2: while not convergent do
3:   when  $M$  vehicle groups are activated, do
4:     for  $j = 1$  to  $M$  do
5:       for  $k = 1$  to  $j - 1$  do
6:         select data batch  $x_j, x_k$ 
7:         exchange information  $h_j, h_k$ 
8:       compute  $\nabla_{\theta_j} \ell(\theta_j, h_j, h_k; y_j)$ 
9:       compute  $\nabla_{\theta_k} \ell(\theta_k, h_j, h_k; y_k)$ 
10:      update model parameters  $\theta_j, \theta_k$  for the two
11:      groups according to Eq. (3)
12:     end for
13:   end for
14: end while
15: return  $\{\theta_m\}$ 

```

V. EVALUATION AND ANALYSIS

In this section, we conduct a set of experiments and discuss the comparison results to evaluate the learning efficiency and reliability for the proposed FL model.

A. Experiment Setup

To investigate the effectiveness of the proposed AS-AFL model under a large-scale distributed ITS, the learning and communication performances are evaluated and analyzed. Specifically, based on three servers with same configuration of two 20-core E5-2660v2 CPUs, 32GB DDR3 RAM, 40G InfiniBand network and 4000MB Ethernet, and Ceph distributed file system, we build a simulated IoT network using Kubernetes tool with Calico CNI plugin to 1000 nodes to conduct the evaluation experiments. Each node is configured with Alpine system using the segmentation strategy including the P2P messaging module, LAN module and WAN module implemented with Golang.

To simulate the dynamic and complex vehicular network with limited computing resources, we actually construct and run 1500 nodes totally. While only 1000 containers will be brought online concurrently at any time. The other 500 nodes will be suspended for scheduling. We comprehensively use Kubernetes Pods and token buckets to limit the network bandwidth. A Pod represents a LAN cluster, inter-Pod bandwidth represents the bandwidth between LAN clusters, and Pod bandwidth is managed by Calico. The bandwidth of the container is limited by the token bucket. Each time the container sends a message, it needs to obtain a token of the message size. If there is no token available in the token bucket, the node must wait. By suspending and starting containers in different Pods, multiple nodes sharing data volumes can achieve the effect of drifting between Pod networks to simulate the entry and exit of nodes between different networks.

In addition, the experiments use the EMNIST dataset for evaluating the learning efficiency. Overall, the dataset contains 344307 training samples and 58646 testing samples separated in 10 categories, where each sample is a 28×28 pixel 8-bit black-and-white image.

B. Evaluation on Learning Efficiency

We first investigate the learning efficiency of the proposed asynchronous aggregation scheme when handling the heterogenous training tasks in the complex network-agnostic environment. The federated aggregation module of the proposed AS-AFL model, is evaluated against the state-of-the-art FL implementations with the purely synchronous update (later referred to as S-FedAvg) that represents the scenario of traditional centralized FL approach, and the purely asynchronous update (later referred to as A-FedAvg). Performances evaluated using the loss and accuracy curve for 200 iterations are illustrated in Fig. 3(a) and (b), respectively.

As shown in Fig. 3, the general trend can be observed in all three implementations where lower losses and higher accuracies can be achieved with longer execution time (i.e., more iterations). However, different approaches take different amount of time to complete 200 iterations of training. In particular, the S-FedAvg that requires to wait for synchronous parameter updates obviously achieves the worst result in terms of training time. The A-FedAvg performs and completed 200 iterations in 957s in a relatively reasonable way. This is because each vehicle may conduct their own

TABLE I
PERFORMANCE COMPARISONS AMONG FIVE DIFFERENT METHODS

Methods	WAN upload (GB)	Avg. WAN Latency (ms)	LAN upload (MB)	Avg. LAN Latency (ms)	Network Throughput (GB)	Convergence Time (hours)	Accuracy (%)
SSGD-FL	120.4	100	n/a	n/a	120.4	579.9	84.54
ASGD-FL	189.5	100	n/a	n/a	189.5	454.8	91.73
EASGD-FL	>>	100	n/a	n/a	>>	>>	-
SGD-Gossip-FL	43.4	100	n/a	n/a	43.4	173.6	89.03
AS-AFL	11.7	100	228.2	20	239.9	46.8	94.03

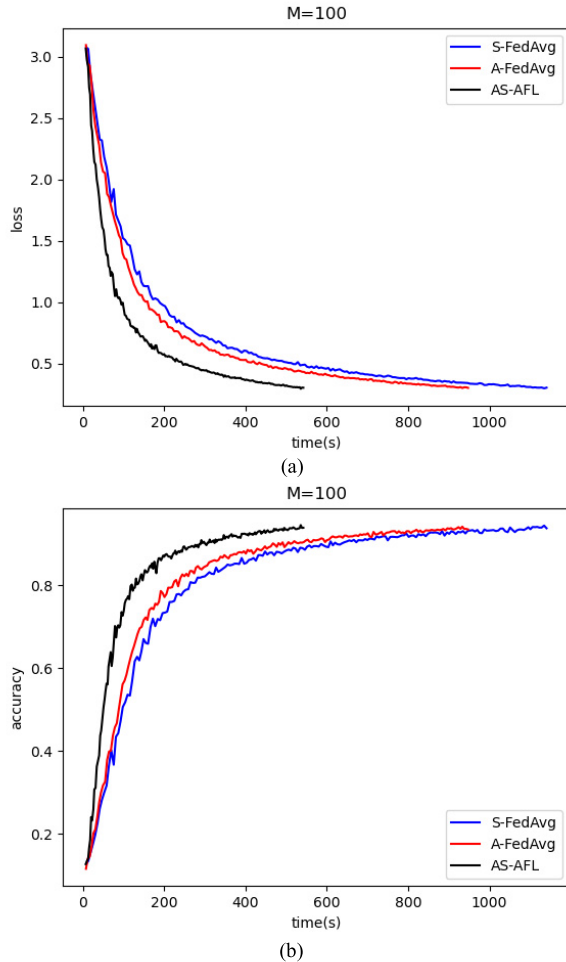


Fig. 3. (a) Loss and (b) Accuracy comparison among AS-AFL, S-FedAvg, and A-FedAvg.

training without waiting for all the parameter updates from other vehicles. However, it still results in some delay to wait for parameter updating to propagate to the entire network, which would not be very suitable in the large-scale distributed system. The proposed AS-AFL achieves the best performance and only takes 542s to complete 200 iterations. This indicates that the meta-learning based segmentation scheme effectively separates similar vehicles into the same group to achieve the efficient synchronous aggregation while allowing asynchronous aggregation across different groups, thus no delay is incurred to wait for other groups to complete and update parameters.

On the other hand, it can be observed obviously that the proposed AS-AFL converges faster than the other two baseline methods. The S-FedAvg achieves the worse result (in time) and only reaches an accuracy of 0.87 and a loss of 0.55. At 542s, the AS-AFL reaches an accuracy of 0.93 and a loss of 0.3, whereas A-FedAvg only reaches an accuracy of 0.88 and a loss of 0.49. Additionally, it is worth noticing that for both A-FedAvg and S-FedAvg, the accuracy and loss curves show more fluctuations, which means their learning processes are not very smooth. In comparison, the proposed AS-AFL, by segmenting nodes into similar groups, is able to achieve a much smoother learning process.

C. Evaluation on Communication Efficiency

To evaluate the effectiveness for the proposed AS-AFL framework, we further design experiments to demonstrate the performance in terms of the communication overhead, network throughput, convergence time, and overall learning accuracy, by comparing the performance of synchronous SGD, asynchronous SGD, and adaptive SGD for a network that consists of over 1000 nodes. The overall hierarchical network topology includes the centralized P2P network and decentralized mesh network. We investigate the gradient descent performance during the training process to figure out how the meta-learning scheme works in the proposed model. Therefore, different gradient descent algorithms, including the synchronous algorithm SSGD, asynchronous algorithm ASGD, hybrid synchronous and asynchronous algorithm EASGD, and adaptive hybrid algorithm SGD-Gossip, are taken into account as the baseline methods. The relationship between multiple parameters is compared through control variables, and the actual performance of the framework is evaluated from multiple dimensions.

The experiment data is obtained by using Prometheus to observe the host indicators and container indicators, and the indicators of all containers in the entire simulated network is recorded at a per second interval. The comparison results are summarized in Table I.

As shown in Table I, the fully synchronous approach SSGD takes the longest time to converge as expected. This is because the model aggregation in SSGD needs to be carried out in a synchronous fashion. The fully asynchronous approach achieves the best accuracy but also takes a long time for all the models to aggregate in an asynchronous fashion. The proposed AS-AFL achieves the best balance of 94.03% accuracy and a great improvement in convergence time in comparison with

the SSGD and ASGD approaches. It is worth noticing that in the EASGD, the communication overhead becomes too large and eventually cannot converge to a stable result.

In addition, the proposed AS-AFL significantly reduces the communication overhead at the WAN level, which achieves less than 10% of the overhead in the SSGD and ASGD cases. Meanwhile, the communication delay with LAN is also considered in the proposed method to reduce the convergence time, which is reflected in the fast convergence time compared with other GD methods. The proposed AS-AFL makes better use of the LAN communication resources and reduces the waste of using unnecessary WAN resources, which is ideal for future large-scale distributed networks in ITS.

VI. CONCLUSION

ITS is soon becoming a reality and shaping our future distributed intelligent services and applications. While advanced communication infrastructure is offering great potentials to interconnect large amounts of vehicles and supporting data collection at an unprecedented scale, existing distributed machine learning techniques are still having difficulties in dealing with large-scale systems that consist of heterogeneous devices and networks in the sustainable computing environment.

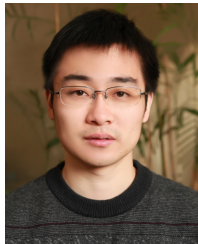
In this paper, the AS-AFL model was proposed and constructed to support future large-scale distributed systems in achieving more effective distributed intelligence. A meta-learning based adaptive segmentation scheme was developed to separate client nodes into multiple edge groups considering the various vehicle and network attributes. A horizontal FL was designed to update parameters within a group of similar client nodes based on the intra-group synchronous aggregation, while a vertical FL was devised to aggregate models across different groups based on the inter-group asynchronous aggregation. The proposed model was evaluated in a distributed ITS application scenario based on an open-source dataset. Evaluation results demonstrated that the proposed AS-AFL could outperform other conventional FL schemes in both learning accuracy and communication efficiency, within a simulated large-scale hierarchical vehicular network that consists of over 1000 nodes.

In future studies, we will go further to investigate different FL schemes and improve the training efficiency with higher precision. Evaluations will be conducted in more complex situations with different traffic scenarios, to enhance the expandability and applicability of our model and algorithm in different real-world scenarios.

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Xiaokang Zhou (Member, IEEE) received the Ph.D. degree in human sciences from Waseda University, Japan, in 2014.

From 2012 to 2015, he was a Research Associate with the Faculty of Human Sciences, Waseda University. He has been a Visiting Researcher with the RIKEN Center for Advanced Intelligence Project (AIP), RIKEN, Japan, since 2017. He is currently an Associate Professor with the Faculty of Data Science, Shiga University, Japan. He has been engaged in interdisciplinary research works in the

fields of computer science and engineering, information systems, and social and human informatics. His recent research interests include ubiquitous computing, big data, machine learning, behavior and cognitive informatics, cyber-physical-social systems, and cyber intelligence and security.

Dr. Zhou is a member of IEEE CS; ACM, USA; IPSJ; JSAI, Japan; and CCF, China.



Wei Liang (Member, IEEE) received the M.S. and Ph.D. degrees in computer science from Central South University, Changsha, China, in 2005 and 2016, respectively.

From 2014 to 2015, he was a Researcher with the Department of Human Informatics and Cognitive Sciences, Waseda University. He is currently with the Key Laboratory of Hunan Province for Mobile Business Intelligence, Hunan University of Technology and Business, Changsha. He has authored or coauthored more than 20 papers at various

conferences and journals. His research interests include information retrieval, data mining, and artificial intelligence.

Dr. Liang is a member of the IEEE Computer Society and the China Computer Federation, China.



Akira Kawai (Member, IEEE) received the B.E., M.E., and Ph.D. degrees in information and computer sciences from Osaka University in 2003, 2005, and 2008, respectively.

He was an Assistant Professor with the Nara Institute of Science and Technology from 2008 to 2013 and the Cybermedia Center, Osaka University, from 2013 to 2014. He is currently an Associate Professor with Shiga University. His research interests include vehicular communication and mobile computing.

Dr. Kawai is a member of IPSJ and IEEE/CS.



Kaoru Fueda received the B.S., M.S., and Ph.D. degrees in mathematics from Kyushu University in 1989, 1991, and 2001, respectively.

He was an Assistant Professor with Kyushu University from 1993 to 2001 and an Associate Professor with Okayama University from 2001 to 2017. He is currently a Professor with Shiga University. His research interests include mathematical statistics.

Dr. Fueda is a member of JSS and JSCS.



Jinhua She (Fellow, IEEE) received the B.S. degree in engineering from Central South University, Changsha, China, in 1983, and the M.S. and Ph.D. degrees in engineering from the Tokyo Institute of Technology, Tokyo, Japan, in 1990 and 1993, respectively. In 1993, he joined the School of Engineering, Tokyo University of Technology, Tokyo, where he is currently a Professor with the Department of Mechanical Engineering. His research interests include the application of control theory, repetitive control, process control, internet-

based engineering education, and assistive robotics. He is a member of the Society of Instrument and Control Engineers, the Institute of Electrical Engineers of Japan, the Japan Society of Mechanical Engineers, and the Asian Control Association. He was a recipient of the International Federation of Automatic Control (IFAC) Control Engineering Practice Paper Prize in 1999 (jointly with M. Wu and M. Nakano).



Kevin I-Kai Wang (Member, IEEE) received the Bachelor of Engineering degree (Hons.) in computer systems engineering and the Ph.D. degree in electrical and electronics engineering from the Department of Electrical and Computer Engineering, University of Auckland, New Zealand, in 2004 and 2009, respectively.

He was a Research Engineer, designing commercial home automation systems and traffic sensing systems from 2009 to 2011. He is currently a Senior Lecturer with the Department of Electrical and Computer Engineering, University of Auckland. His current research interests include wireless sensor network based ambient intelligence, pervasive healthcare systems, human activity recognition, behavior data analytics, and bio-cybernetic systems.